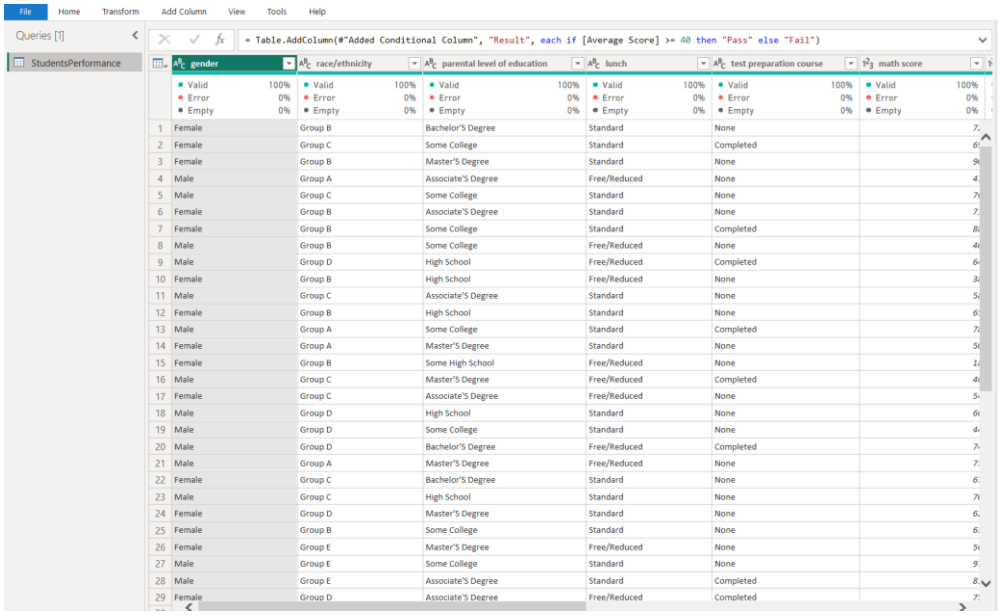
WEEK 7 REPORT **(BY:GEETHASHINI V S)**

1. Objective

The objective of Week 7 was to clean, transform, and prepare the Student Performance dataset using Power BI. I created calculated columns, assigned grades, identified pass/fail status, and unpivoted the subject scores for further analysis.

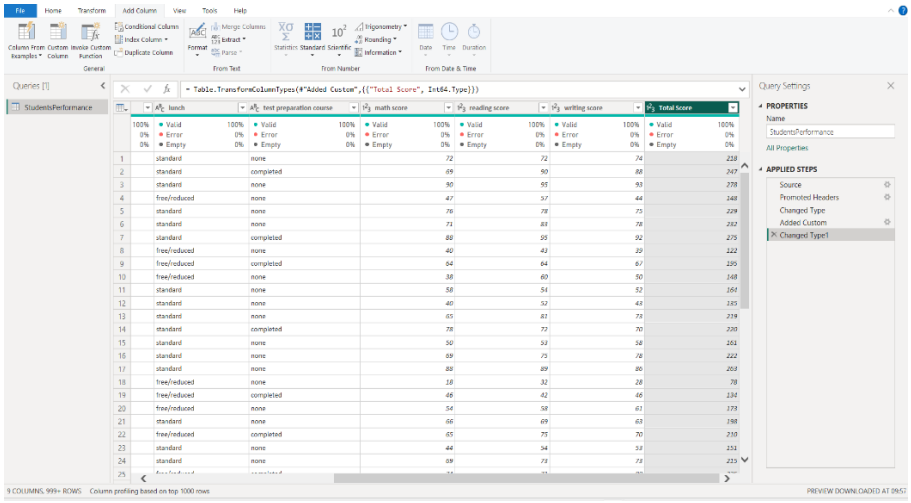
2. Importing the Dataset

I imported the Student Performance dataset into Power BI Desktop. I opened Power Query Editor to inspect the data and prepare it for analysis.



4. Data Cleaning and Data Types

I checked the dataset for missing values, errors, and duplicate records. I also verified that the columns had appropriate data types, such as text for categorical columns and whole numbers for marks.



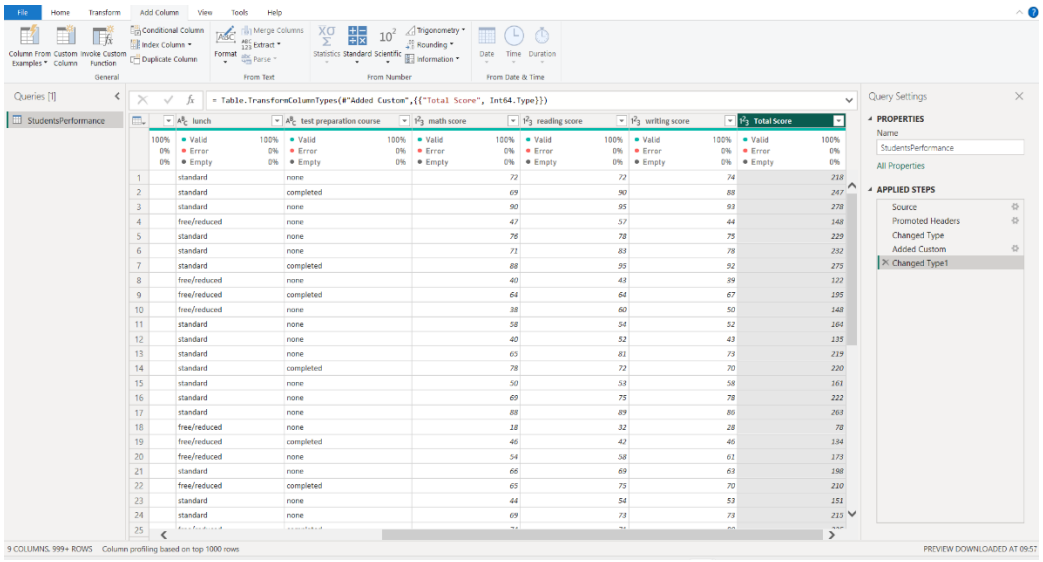
5. Creating the Total Score Column

I created a custom column named **Total Score** by adding the marks obtained in Mathematics, Reading, and Writing.

Formula:

[math score] + [reading score] + [writing score]

This column calculates the total marks obtained by each student across the three subjects.



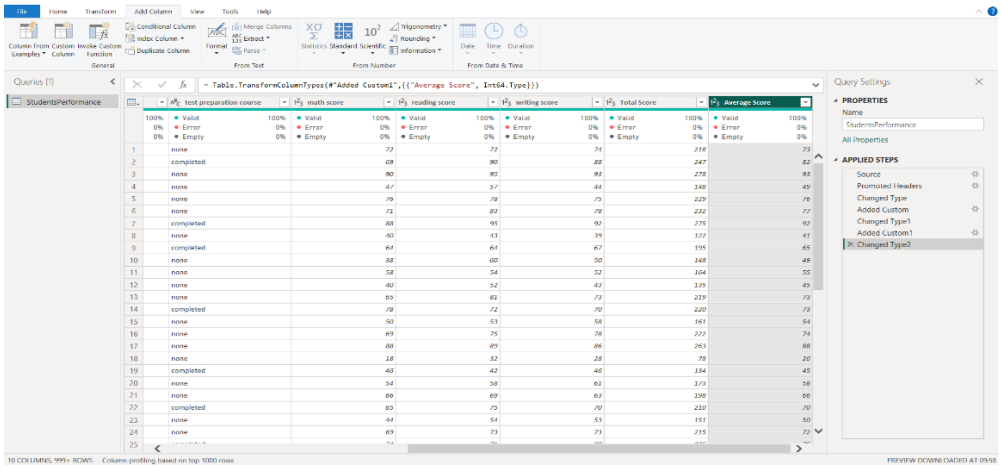
6. Creating the Average Score Column

I created another custom column named **Average Score** by dividing the Total Score by 3. I rounded the result to the nearest whole number.

Formula:

Number.Round([Total Score] / 3, 0)

This column helps to understand each student's average performance across the three subjects.

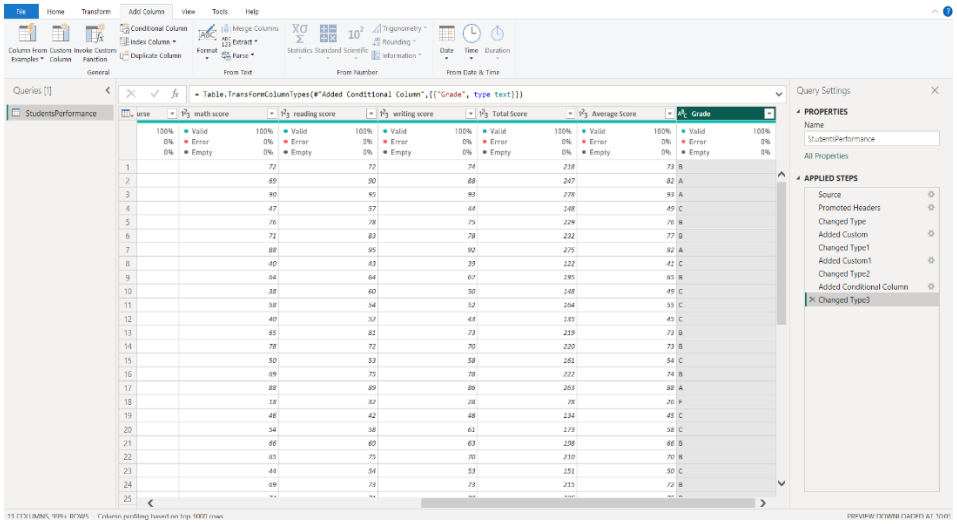


7. Creating the Grade Column

I created a conditional column named **Grade** based on the Average Score. The following grade boundaries were used for this exercise:

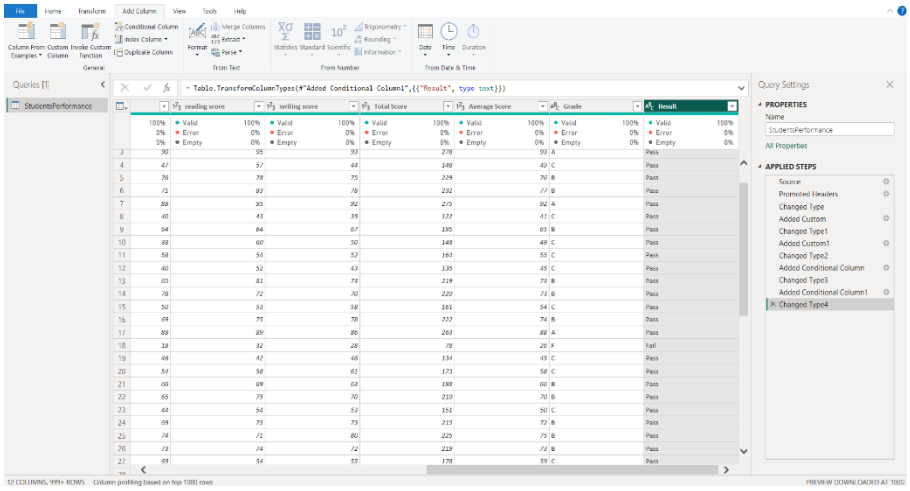
- Average Score of 80 or above – A
- Average Score of 60 or above – B
- Average Score of 40 or above – C
- Below 40 – F

These boundaries were used as the grading rule for this exercise.



8. Creating the Result Column

I created a conditional column named **Result** based on the Average Score. Students with an Average Score of 40 or above were marked as Pass, while students scoring below 40 were marked as Fail.

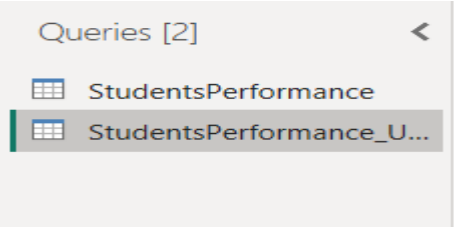


9. Duplicating the Table and Unpivoting Subject Scores

I duplicated the original student dataset and renamed the duplicate table to StudentsPerformance_Unpivot. I selected the Mathematics, Reading, and Writing score columns and used the **Unpivot Columns** option in Power Query.

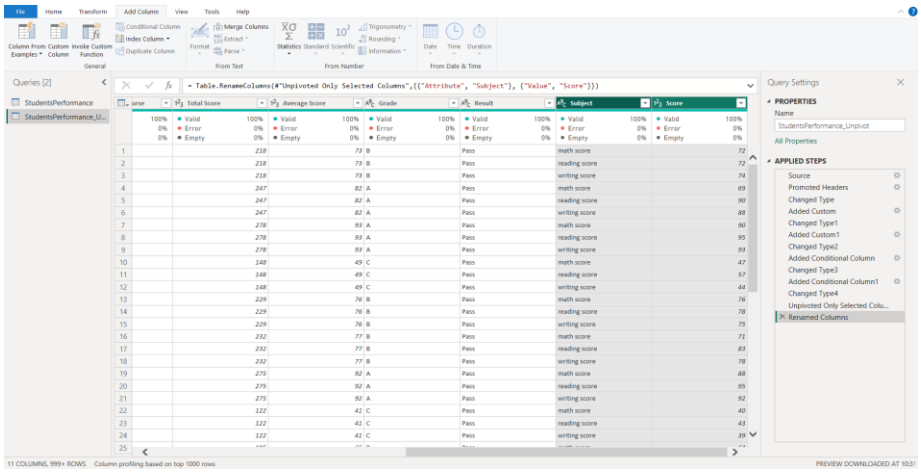
I renamed the resulting Attribute column to **Subject** and the Value column to **Score**. I retained the other student information columns in the table.

This transformation converted the three subject-score columns into rows, making the data easier to analyse by subject.





Subject	Score
Valid 100%	Valid 100%
Error 0%	Error 0%
Empty 0%	Empty 0%
math score	72
reading score	72
writing score	74
math score	69
reading score	90
writing score	88
math score	90
reading score	95
writing score	93
math score	47
reading score	57
writing score	44
math score	76
reading score	78
writing score	75
math score	71
reading score	83
writing score	78
math score	88
reading score	95
writing score	92
math score	40
reading score	43
writing score	39



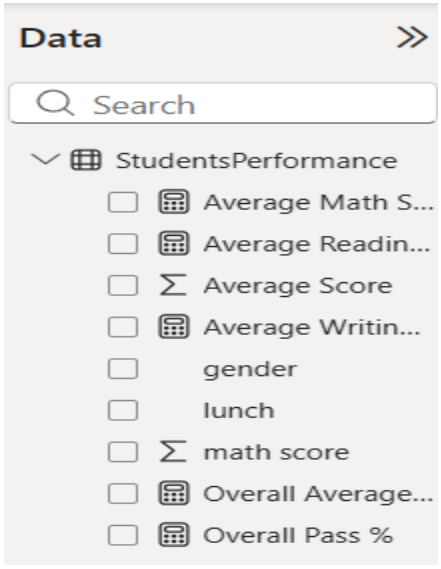
10. Applying Changes and Creating Measures

After completing the data transformations, I selected **Close & Apply** to load the changes into Power BI Desktop.

I then created DAX measures on the original student table to calculate:

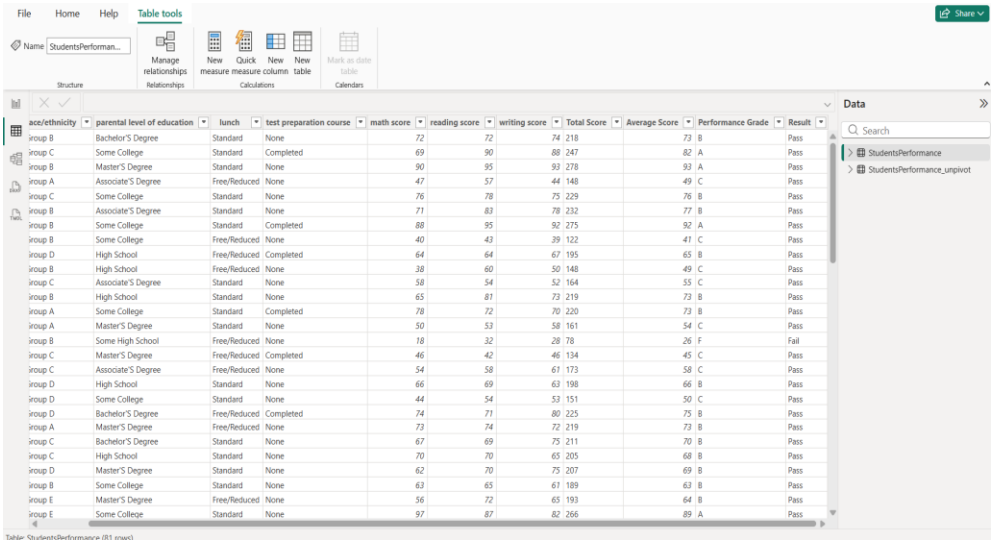
- Average Math Score
- Average Reading Score
- Average Writing Score
- Overall Average Score
- Overall Pass Percentage

These measures help summarize student performance and can be used in Power BI visuals.



11. Validation

I checked the transformed data to ensure that the Total Score and Average Score calculations were correct. I also reviewed the Grade and Pass/Fail columns and verified that the unpivoted table contained the Subject and Score columns along with the other student details.



12. Conclusion

In Week 7, I learned how to import and clean a dataset, change data types, create custom and conditional columns, duplicate a table, and unpivot data using Power Query. I also learned how to create DAX measures for analysing student performance. These activities helped me prepare the dataset for dashboard creation and further analysis in Week 8.